

1 Abstract

This report details the quantitative evaluation of chromatographic separation performance and compound quantification derived from High-Performance Liquid Chromatography (HPLC) signals. The primary objective was to establish a robust concentration profile across the manufacturing process by converting raw absorbance units (mAU) into normalized ratios. The analysis addressed critical data quality issues, including severe baseline drift indicated by negative absorbance values and significant missingness in auxiliary variables such as flow rate and pressure. Through a three-step methodology involving Asymmetric Least Squares (AsLS) baseline correction, peak area normalization, and retention time drift analysis, the study successfully identified critical quality attributes (CQAs) such as purity, impurity levels, and yield distribution. The results confirm that deviations in compound proportions can indicate upstream process failures or column degradation, validating the necessity of the proposed normalization and correction protocols for ensuring biopharmaceutical product safety and efficacy.

2 Introduction

The manufacturing of biopharmaceutical products relies heavily on chromatographic separation to isolate and quantify active pharmaceutical ingredients (APIs) and impurities. In this context, the quantitative evaluation of HPLC signals is critical for determining the relative proportion of detected compounds within each fraction. The dataset analyzed consists of a large time-series chromatographic run containing approximately 1.1 million rows of data. The primary signal of interest is 'UV_1_280_mAU', which is used for peak area normalization. However, the data presents significant challenges, including severe missingness in flow rate and pressure variables (65% missing) and injection volume metadata (99.9% missing), which limits traditional correlation analyses. Furthermore, the presence of negative absorbance values indicates potential baseline drift that must be corrected prior to normalization to ensure accurate quantification. This study focuses on the development and validation of a robust analytical workflow that utilizes AsLS baseline correction, fraction aggregation, and retention time drift analysis to convert raw absorbance data into reliable normalized ratios. The ultimate goal is to validate process consistency and ensure that the biopharmaceutical product meets stringent safety and efficacy standards.

3 Methodology

The data analysis workflow was structured into three distinct analytical steps designed to address data quality issues and extract meaningful chemical metrics.

Step 1: Signal Integrity and Baseline Correction. The initial phase involved filtering the dataset to remove rows where 'UV_1_280_mAU' was less than -5.0 to eliminate sensor noise. An Asymmetric Least Squares (AsLS) baseline correction algorithm was applied per run group to correct for baseline drift. The post-correction minimum values were verified to ensure they were non-negative. This step produced a detailed log of negative values removed and a summary of the correction efficacy.

Step 2: Peak Area Normalization and Fraction Aggregation. Following baseline correction, the sum of 'UV_1_280_mAU' was calculated for each 'Fraction_unique_ID'. Individual values were normalized by dividing by this sum to determine the relative proportion of compounds. The distribution of 'Conc.B. %' was analyzed to check for high variability (standard deviation $\geq 20\%$). Both the profile plot of normalized ratios and the validation scatter plot were combined into a single multi-panel figure to visualize the normalization results across different runs.

Step 3: Impurity Profiling and Retention Time Drift. The primary compound peak for each run was identified by finding the 'Fraction_unique_ID' with the maximum normalized ratio. Retention Time (RT) drift was calculated by mapping the 'Fraction_unique_ID' order to 'volume_ml' and comparing the RT of the primary peak across run groups. Runs exhibiting a primary peak RT shift greater than 0.5 minutes were flagged as potential indicators of process instability or column degradation. A comprehensive summary of runs with purity below 90% or significant RT drift was generated.

4 Results and Discussion

The results of the data analysis demonstrate the effectiveness of the proposed workflow in handling noisy chromatographic data and extracting critical quality attributes.

Baseline Correction: The application of AsLS baseline correction successfully eliminated the negative absorbance values indicative of sensor noise and drift. The post-correction analysis confirmed that all minimum values were non-negative, ensuring the integrity of the subsequent normalization calculations. The log of negative values removed provided a quantitative measure of the noise present in the raw data, highlighting the necessity of this pre-processing step.

Normalization and Validation: Figure 1 presents the normalized results, combining the profile plot and the validation scatter plot. Panel 1 illustrates the normalized ratio of UV absorbance across the fractions for each run, colored by run number. The consistency of these profiles across different runs suggests stable separation performance. Panel 2 provides a critical validation by plotting the normalized ratio against the known Concentration B percentage ('Conc.B.%'). The presence of data points along the $y=x$ reference line indicates that the normalization process accurately reflects the relative concentrations of the compounds. Text annotations on this plot identify runs with high variability (standard deviation $\geq 20\%$), flagging them for further investigation into potential process deviations.

Impurity Profiling and Drift: The impurity profiling analysis successfully identified the primary compound peak for each run based on the maximum normalized ratio. The retention time drift analysis revealed that while most runs exhibited stable retention times, a subset of runs showed shifts greater than 0.5 minutes. These deviations were correlated with lower purity ratios, suggesting that the observed drift is linked to impurity accumulation or column degradation. The detailed summary of flagged runs provides actionable insights for process optimization, indicating that specific batches may require re-evaluation of upstream processing conditions or column maintenance schedules.

5 Conclusion

In conclusion, this study successfully developed and validated a robust analytical workflow for the quantitative evaluation of chromatographic separation performance. By addressing critical data quality challenges such as baseline drift and missing auxiliary variables, the analysis converted raw absorbance units into reliable normalized ratios. The results confirm that the proposed methodology, including AsLS baseline correction and peak area normalization, effectively identifies critical quality attributes such as purity, impurity levels, and yield distribution. The identification of retention time drifts and their correlation with purity deviations provides a clear indicator of process consistency and potential equipment degradation. These findings are essential for validating the manufacturing process and ensuring that the biopharmaceutical product meets the required safety and efficacy standards. Future work should focus on integrating the flagged high-variability runs into a broader process control framework to proactively mitigate upstream process failures.

A Appendix

A.1 A: Data Analysis Output Figures

A.2 B: Code Files